UNIVERSITY COLLEGE BIRMINGHAM

Network Segmentation and Security Implementation in a Medium-Sized Enterprise Environment

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Part 1: Theoretical Tasks

The benefits of segmenting a network into multiple VLANs and Subnets

Understanding VLANs and Subnets

The process of dividing a network into subnets (sub-networks) which in turn are divided into VLANs (Virtual Local Area Networks), is always used as the first step to maintenance and security of modern computer networks. Layer 2 technology allows the network administrator to logically group network devices to the function or department, instead of physical location. On the other hand, subnetting is done in Layer 3 and breaks down vast IP networks into a series of smaller IP network segments (Okoh et al., 2024). Similar to VLANs, creating subnets also allows creating boundaries in a network thereby allowing control, efficiency and security.

Enhancing Network Security through Logical Separation

Networks are segmented mainly to increase their security. Devices in isolated networks formed by VLANs and subnets are limited in their communication unless it is permitted expressly. If a device on a flat network is compromised, it can effortlessly spread malware or illegally gain access to important data (Okoh et al., 2024). If the network is separated into parts, there is less chance for lateral movement. Unless routing and access permissions are set up, employees from one department cannot access the resources or servers in another department. As a result of being separated, critical systems are protected from threats that may come from inside or outside the network.

Improving Performance by Reducing Broadcast Traffic

Segmentation is useful for controlling the amount of information being sent. Having many devices on a big LAN results in all the computers receiving and processing broadcast packages and it may overload the network. With VLANS, broadcasts are limited to their respective VLANs. Similarly, subnets avoid sending superfluous broadcasts through routers and only handle traffic between subnets when it is required (Al-Khaffaf and Al-Hamiri, 2021). By using this method, all devices are allocated bandwidth only for messages meant for them which makes VoIP, online meetings and other real-time tools operate more smoothly.

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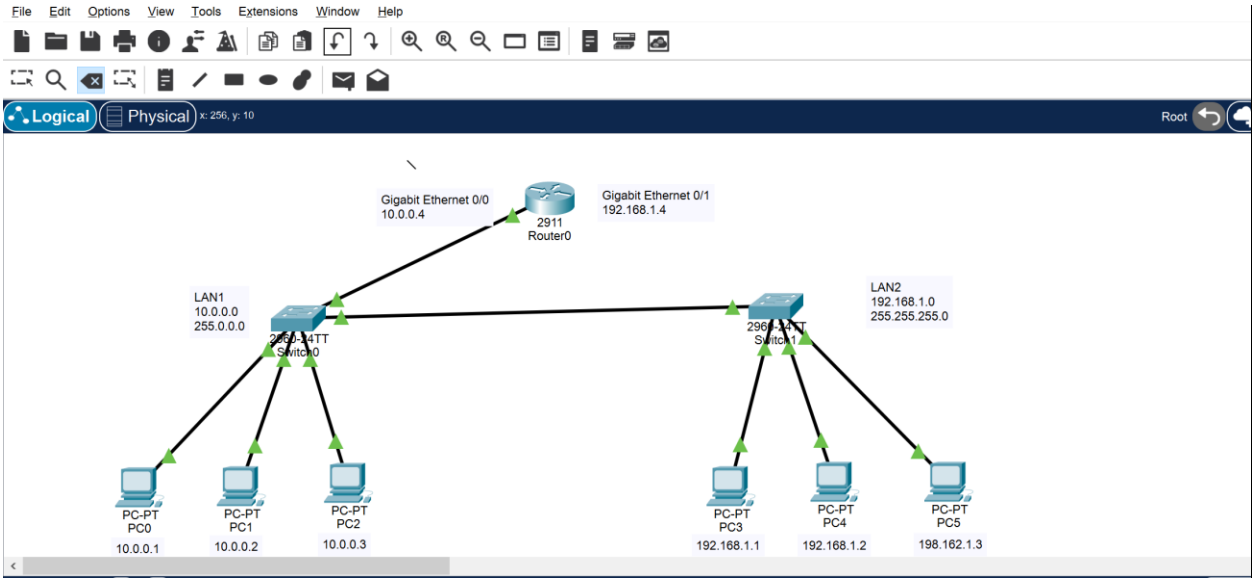

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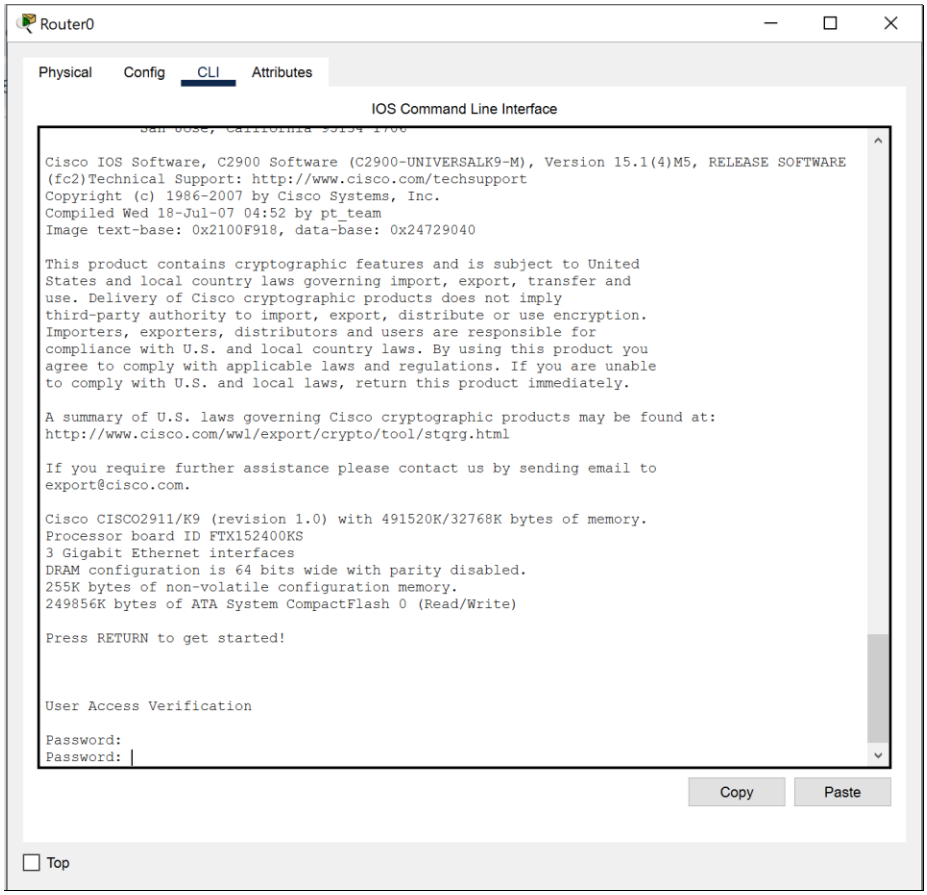
```
Switch>enable
Password:
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name SALES
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 20
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#interface range fa0/1 - 3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface range fa0/4 - 6
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface fa0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#
Switch(config)#end
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (10), with
Switch FastEthernet0/1 (1).

Switch#
%SYS-5-CONFIG_I: Configured from console by console
```



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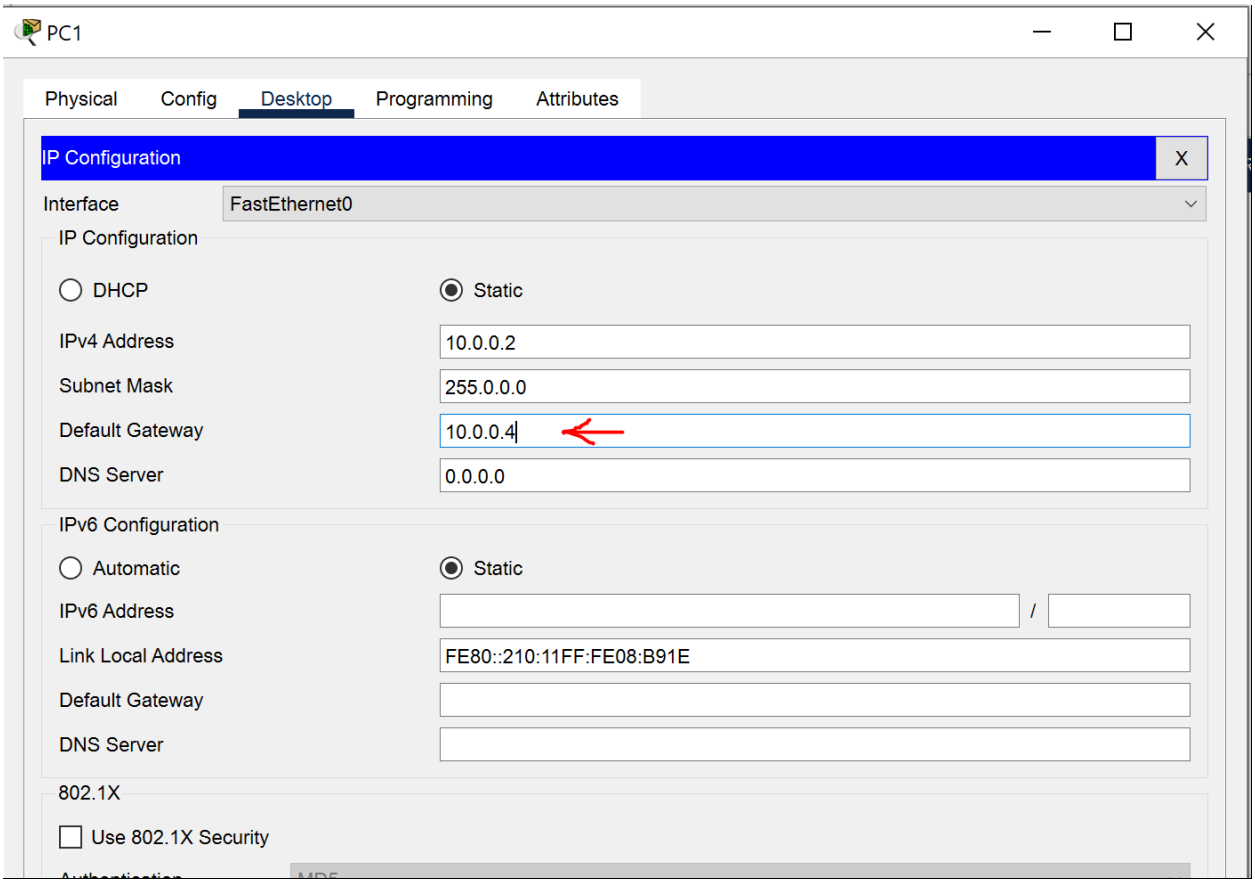
PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.0.0.2
Subnet Mask
255.0.0.0
Default Gateway
10.0.0.4
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::210:11FF:FE08:B91E
Default Gateway
DNS Server
802.1X
Use 802.1X Security

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PC1

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.1: bytes=32 time=1ms TTL=127
Reply from 192.168.1.1: bytes=32 time<1ms TTL=127
Reply from 192.168.1.1: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
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